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QUESTION 8 • UNIT 2 – TOOLS & FRAMEWORKS

Write short note on PyTorch and Google Colab.

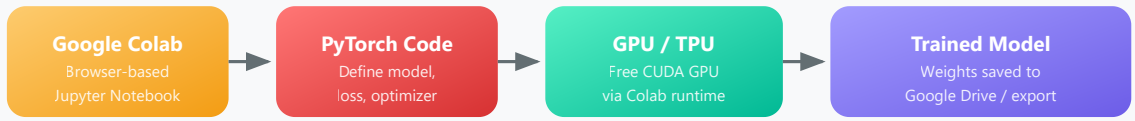
📝 5 Marks

PyTorch & Google Colab

Deep learning framework + cloud notebook platform — with diagrams

Overview — How They Work Together

Figure 1 — PyTorch + Colab Workflow



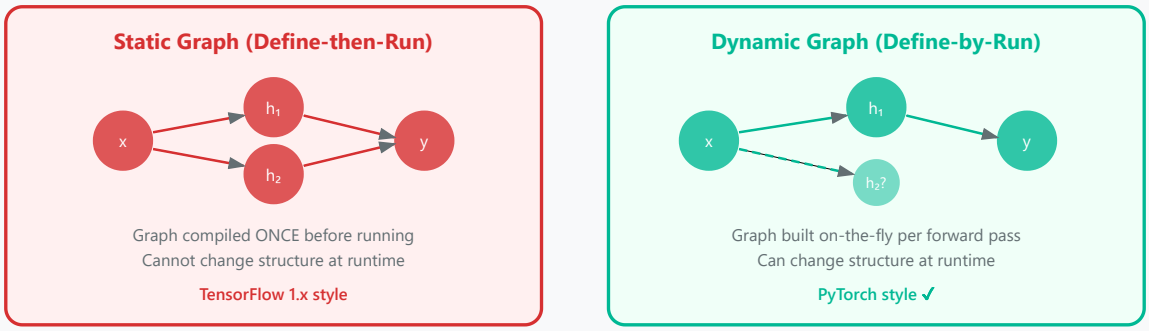
Write PyTorch in Colab → run on free GPU → save trained model

Colab provides the cloud environment; PyTorch provides the deep learning framework; GPU accelerates training.

PyTorch

PyTorch is an open-source machine learning framework developed by **Facebook's AI Research (FAIR)** lab. It is widely used for deep learning due to its dynamic computation graph which allows real-time changes to network architecture during training.

Figure 2 — Static Graph (TensorFlow) vs. Dynamic Graph (PyTorch)



PyTorch's dynamic graph allows adding/removing branches per input — ideal for research and variable-length sequences.

Key Features of PyTorch

Dynamic Computation Graph

Graph is built at runtime — enables flexibility and easy debugging.

Autograd (Auto Differentiation)

Automatically computes gradients for backpropagation using the `autograd` module.

GPU Acceleration (CUDA)

Models run on NVIDIA GPUs via CUDA, achieving massive speedups.

Pythonic API

Integrates seamlessly with Python — uses familiar NumPy-like syntax.

Rich Ecosystem

torchvision, torchtext, torchaudio — libraries for CV, NLP, audio.

Large Community

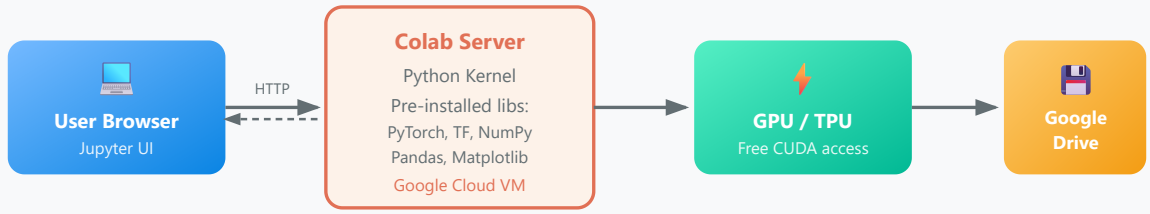
Extensive documentation, tutorials, and pre-trained models (HuggingFace, etc.).

Google Colab

Google Colab (Google Collaboratory) is a **free, cloud-based platform** for writing and executing Python code in a Jupyter Notebook environment. No installation is required —

everything runs in the browser.

Figure 3 — Google Colab Architecture



Colab runs on Google Cloud VMs — code is written in a browser, executed on a remote GPU/TPU, and files saved to Google Drive.

Key Features of Google Colab

Free GPU & TPU Access

Free access to NVIDIA GPUs and Google TPUs — ideal for training deep learning models.

Cloud-Based, No Setup

No installation required — runs entirely in the browser on Google Cloud infrastructure.

Google Drive Integration

Mount Google Drive directly to save and access datasets and notebooks.

Real-Time Collaboration

Multiple users can work on the same notebook simultaneously, like Google Docs.

Pre-installed Libraries

TensorFlow, PyTorch, NumPy, Pandas, Matplotlib — ready to use out of the box.

Interactive Notebooks

Combines code, rich text, images, and output — ideal for tutorials and presentations.

PyTorch vs. Google Colab — Comparison

Aspect	PyTorch	Google Colab
Type	Deep learning framework	Cloud notebook platform
Purpose	Build & train neural network models	Provide environment to write & run code
Created by	Facebook AI Research (FAIR)	Google
Language	Python (C++ backend)	Python (Jupyter-based)
Hardware	Uses local GPU (CUDA) or cloud GPU	Provides free cloud GPU/TPU
Installation	Requires local install (`pip install torch`)	No install — browser only
Best for	Research, model development, deployment	Learning, prototyping, collaboration